Ogbonna Arinze

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SUMMARY

Senior Full-Stack Engineer with 8 years of professional experience building high-impact web applications across healthcare, fintech, civic tech, and enterprise B2B SaaS environments. Proven expertise spanning **React, Redux, TypeScript, Node.js**, and **NestJS**, with deep experience in **MongoDB**, **AWS**, and production API development. Track record of delivering scalable platforms serving thousands of users, driving architecture modernization, and establishing robust testing practices with **Jest, Cypress**, and **Playwright**. Effective communicator in international, English-speaking teams with a strong sense of ownership in lean engineering organizations.

CORE COMPETENCIES

- **Frontend:** React, Redux, Next.js, TypeScript, JavaScript (ESNext), Vue.js, Tailwind CSS, Radix UI, Ant Design, and Storybook.
- **Backend:** Node.js, NestJS, REST APIs, MongoDB, PostgreSQL, Prisma, JWT authentication, and role-based access control.
- **Testing & Quality:** Jest, Cypress, Playwright, Allure Reports, Storybook, ESLint, Prettier, TDD/BDD, and CI-driven QA.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker, GitHub Actions, CI/CD pipelines, Nx monorepo orchestration, and Vercel.
- **Architecture:** Scalable component systems, design systems, monorepos (Nx, pnpm workspaces), SSR/SSG, real-time streaming (SSE, EventSource), and performance optimization.
- **AI & Data:** RAG pipelines, OpenAI API, LangChain, vector search (pgvector, Chroma), embeddings, and data visualization (Recharts, react-chartjs).
- **Leadership:** Technical architecture reviews, mentorship, Agile/Scrum, product-minded development, and cross-functional collaboration.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)

Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React** Chrome extension, optimizing workflows for clinical site integrations and enhancing adoption across users.
- Maintained and extended the existing **Vue.js** codebase during the migration period, fixing bugs and shipping new features in parallel with the **React** rebuild, ensuring zero disruption for end users.
- Co-developed a full-stack **Next.js** chat widget powered by a RAG system using **OpenAI GPT models**, **LangChain**, Vercel AI SDK, and Chroma vector database, enabling users to query their data in real time via a conversational interface.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Implemented **JWT-based authentication** and **role-based access control** in Next.js with **TypeScript**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Optimized application performance by implementing lazy loading, code splitting, and CDN asset delivery, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Increased release reliability by integrating **Playwright** end-to-end tests into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Architected a reusable design system in **React** and Radix UI with Tailwind CSS, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and Enzyme testing frameworks, leading to a **20% decrease** in bug reports and establishing a culture of automated testing.
- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices including precise user story estimation and sprint planning.

- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React**, **Redux**, **RxJS** for real-time data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design** UI, streamlining management of user roles and permissions.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to Heroku and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance, code reviews, and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app (22 chart types with Recharts), Next.js admin dashboard, **NestJS** document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with pnpm workspaces.
- Built the **NestJS** backend with **PostgreSQL 16** and **Prisma v7**, implementing SSE streaming for real-time chat responses with 15s keepalive heartbeats, phone OTP authentication, and role-based access control across user management, analytics, and admin features.
- Designed a multi-agent AI system using Mastra with 5 specialized agents and a three-stage intent classification pipeline, powered by a hybrid RAG pipeline combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and Cohere reranking.
- Configured **CI/CD** with **GitHub Actions** for automated builds and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown. Integrated Traefik for reverse proxy routing.
- Integrated Langfuse for LLM observability, Infisical for secrets management, and **Playwright** E2E tests across 4 test projects (web auth, dashboard auth, desktop Chrome, mobile Pixel 5).

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*